thm:abstract-p-isom in model-theory/lindstrom

The Open Logic Project contributors

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Editorial note. All mathematical statement and proof text is reproduced verbatim from the Open Logic Project, unit model-theory, file ls-property.tex, fixed at commit 1e960beff9ed7835bf3e3f1335e21af3439cd107. Only the packaging (the adapter preamble and the standalone wrapper) is this module's.

Editorial note. Numbering inside this standalone excerpt is editorial, not the Open Logic Project print numbering.

Editorial note. The adapter preamble loads the project's OWN style layer verbatim (sty/open-logic-formulas.sty, sty/open-logic-tokenize.sty, sty/open-logic-selective.sty and open-logic-config.sty, including its tag truth assignment), so the author's text-token shorthand, !A math metavariables, rule names (Intro, Elim, FalseInt) and the tag conditionals resolve exactly as in the book. Three substitutions are forced by the pinned toolchain and are recorded: isomath.sty is absent so \mathsfit is \mathsf; xpunctuate.sty is absent and unused so its RequirePackage is dropped; thmtools/ntheorem are absent so the environments are amsthm with the project's family numbering. The proof trees are the project's own bussproofs-extra.

Editorial note. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof at the pinned line span. Original candidate key: model-theory/lindstrom|ls-property.tex|thm:abstract-p-isom.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites. The author's own ollabel is kept unprefixed (not part:chapter:section) so those pointers resolve; the label text itself is the author's own argument.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own olref pointers. Its prerequisites are the earlier results of the same pinned Open Logic Project commit that the author cites there.

Source reference. Original definition 0000: sources/partial-iso.tex, source lines 36–36. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub. No bibliography citation is involved.

Source reference. Original theorem 0000: sources/partial-iso.tex, source lines 52–52. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub. No bibliography citation is involved.

Theorem 0.1. *Suppose $\langle L, \models_L \rangle$ is a normal logic with the Löwenheim–Skolem property. Then any two structures that are partially isomorphic are elementarily equivalent in $\langle L, \models_L \rangle$.*

Proof. Suppose $\mathfrak{M} \simeq_p \mathfrak{N}$, but for some α also $\mathfrak{M} \models_L \alpha$ while $\mathfrak{N} \not\models_L \alpha$. By the Isomorphism Property we can assume that $|\mathfrak{M}|$ and $|\mathfrak{N}|$ are disjoint, and by the Expansion Property we can assume that $\alpha \in L(\mathcal{L})$ for a finite language $\mathcal{L}$. Let $\mathcal{I}$ be a set of partial isomorphisms between $\mathfrak{M}$ and $\mathfrak{N}$, and with no loss of generality also assume that if $p \in \mathcal{I}$ and $q \subseteq p$ then also $q \in \mathcal{I}$.

$|\mathfrak{M}|^{<\omega}$ is the set of finite sequences of elements of $|\mathfrak{M}|$. Let S be the ternary relation over $|\mathfrak{M}|^{<\omega}$ representing concatenation, i.e., if $\mathbf{a}, \mathbf{b}, \mathbf{c} \in |\mathfrak{M}|^{<\omega}$ then $S(\mathbf{a}, \mathbf{b}, \mathbf{c})$ holds if and only if $\mathbf{c}$ is the concatenation of $\mathbf{a}$ and $\mathbf{b}$; and let T be the ternary relation such that $T(\mathbf{a}, b, \mathbf{c})$ holds for $b \in M$ and $\mathbf{a}, \mathbf{c} \in |\mathfrak{M}|^{<\omega}$ if and only if $\mathbf{a} = a_1, \ldots a_n$ and $\mathbf{c} = a_1, \ldots a_n, b$. Pick new 3-place predicate symbols P and Q and form the structure $\mathfrak{M}^*$ having the universe $|\mathfrak{M}| \cup |\mathfrak{M}|^{<\omega}$, having $\mathfrak{M}$ as a substructure, and interpreting P and Q by the concatenation relations S and T (so $\mathfrak{M}^*$ is in the language $\mathcal{L} \cup \{P, Q\}$).

Define $|\mathfrak{N}|^{<\omega}$, S', T', P', Q' and $\mathfrak{N}^*$ analogously. Since by hypothesis $\mathfrak{M} \simeq_p \mathfrak{N}$, there is a relation I between $|\mathfrak{M}|^{<\omega}$ and $|\mathfrak{N}|^{<\omega}$ such that $I(\mathbf{a}, \mathbf{b})$ holds if and only if $\mathbf{a}$ and $\mathbf{b}$ are isomorphic and satisfy the back-and-forth condition of 0000. Now, let $\mathfrak{M}$ be the structure whose domain is the union of the domains of $\mathfrak{M}^*$ and $\mathfrak{N}^*$, having $\mathfrak{M}^*$ and $\mathfrak{N}^*$ as substructures, in the language with one extra binary predicate symbol R interpreted by the relation I and predicate symbols denoting the domains $|\mathfrak{M}|^*$ and $|\mathfrak{N}|*$.

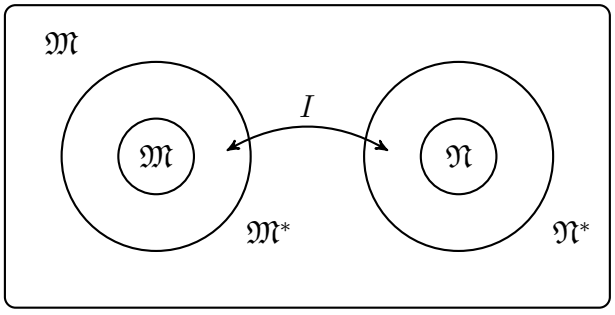

Figure 1: The structure $\mathfrak{M}$ with the internal partial isomorphism.

The crucial observation is that in the language of the structure $\mathfrak{M}$ there is a *first-order* sentence θ_1 true in $\mathfrak{M}$ saying that $\mathfrak{M} \models_L \alpha$ and $\mathfrak{N} \not\models_L \alpha$ (this requires the Relativization Property), as well as a *first-order* sentence θ_2 true in $\mathfrak{M}$ saying that $\mathfrak{M} \simeq_p \mathfrak{N}$ via the partial isomorphism I. By the Löwenheim–Skolem Property, θ_1 and θ_2 are jointly true in an enumerable model $\mathfrak{M}_0$ containing partially isomorphic substructures $\mathfrak{M}_0$ and $\mathfrak{N}_0$ such that $\mathfrak{M}_0 \models_L \alpha$ and $\mathfrak{N}_0 \not\models_L \alpha$. But enumerable partially isomorphic structures are in fact isomorphic by 0000, contradicting the Isomorphism Property of normal abstract logics. □